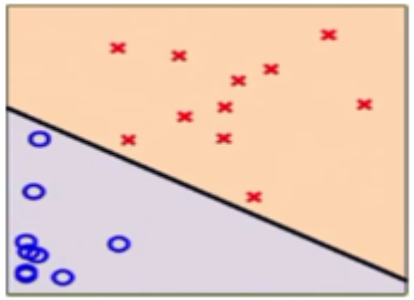
Classification

Motivation

Data

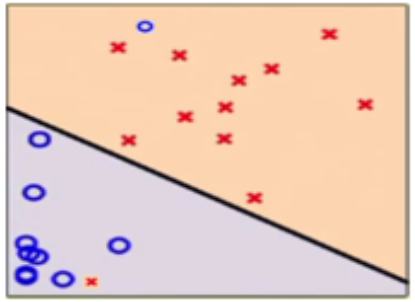
Linearly separable



Without noise

Perceptron

PLA



With noise

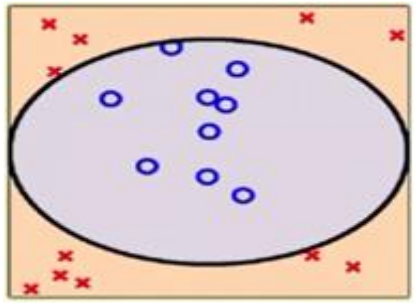
Adaline

Delta rule

Perceptron

Pocket

Nonlinearly separable



Nonlinear Transformation

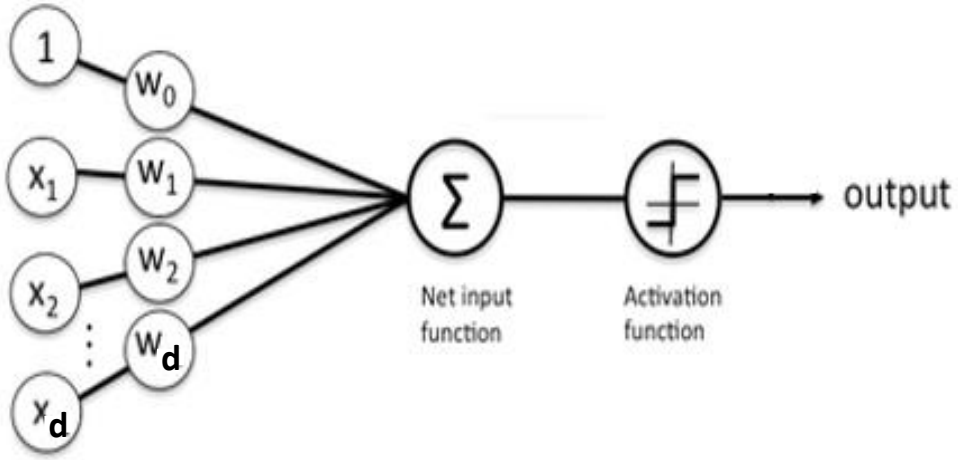
Perceptron

The perceptron hypothesis is:

$$h\,(x) = \begin{cases} +1 & si \quad w^T x \geq b \\ -1 & si \quad w^T x < b \end{cases} \text{ où } w \in R^d$$

$$h\,(x) = sign(w^T x) = \begin{cases} +1 & si \quad w^T x \geq 0 \\ -1 & si \quad w^T x < 0 \end{cases} \text{ où } w \in R^{d+1}$$

d is the dimension of the input space.



Perceptron

We have two sets N and P:

$$\begin{cases} if\ x \in P & \rightarrow \quad y = +1 \\ if\ x \in N & \rightarrow \quad y = -1 \end{cases}$$

Objective:

We look for w capable of absolutely separating the two sets N and P:

$$P = \text{open positive half space}$$

$$\mathrm{N} = \text{open negative half space}$$

To simplify the visualization of the algorithm, we are going to take $d = 2$.

So:

$$x = (x_1, x_2) \text{ et } \mathrm{w} = (w_1, w_2)$$

$$h(x) = \langle w, x \rangle = w_1 x_1 + w_2 x_2$$

Perceptron

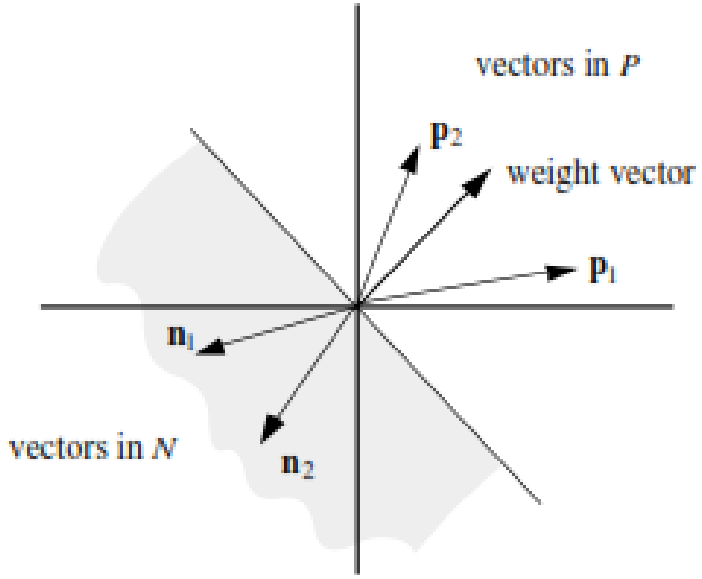
Notice that:

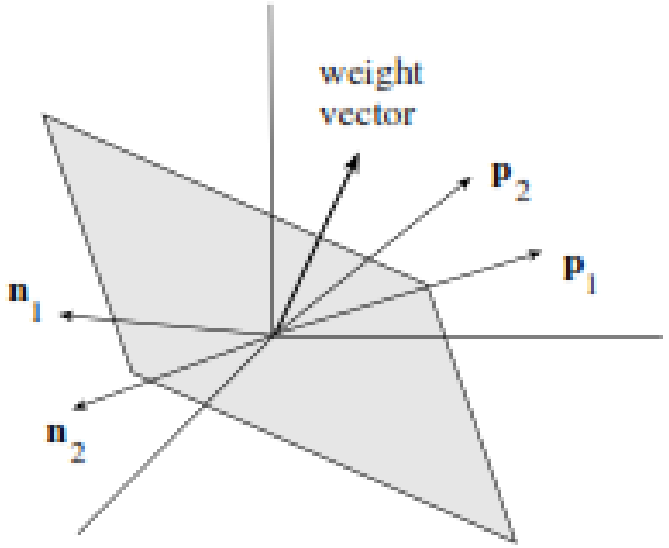
$$w_1 x_1 + w_2 x_2 = 0$$

Is the equation of a plane. And the vector normal to this plane is the weight vector $\mathrm{w} = (w_1, w_2)$.

We can visualize the linear representation in two different spaces:

- Input space: $x = (x_1, x_2)$ etr $\mathrm{w} = (w_1, w_2)$
- Extended input space: $x = (1, x_1, x_2)$ et $\mathrm{w} = (w_0, w_1, w_2)$





Perceptron

Perceptron learning algorithm

Input: $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and w_0.
Output: w^*, t and $L_S(w^*)$
Start: $w \leftarrow w_0$ and $\mathrm{t} \leftarrow 0$

Compute: $L_S(w) = \frac{1}{n}\sum_{i=1}^{n} 1_{[w^T x_i \neq y_i]}$

```
While (L_S(w)! = 0) :
  for i = 1, ..., n:
      if signe(w^T x_i).y_i < 0
        w ← w + y_i x_i
        t ← t + 1
      endif
  endfor
  compute L_S(w)
endWhile
```

Return $w^* \leftarrow \mathrm{w}$, $L_S(w^*)$ and t.

end

Perceptron

Objective: Reformulation

We should have that:

$$\begin{cases} \forall x \in P \ , & \langle w, x \rangle \geq 0 \\ \forall x \in \mathrm{N} \ , & \langle w, x \rangle < 0 \end{cases}$$

We know that:

$$\langle w, x \rangle = \|w\|.\|x\| \cos(w, x) = \|w\|.\|x\| \cos(\alpha)$$

$$\cos(\alpha) = \frac{\langle w, x \rangle}{\|w\|.\|x\|}$$

$$\alpha = \arccos\left(\frac{\langle w, x \rangle}{\|w\|.\|x\|}\right)$$

Perceptron

Notice that:

$$if\ \langle w, x\rangle < 0\ \Longrightarrow \cos(\alpha) < 0 \Longrightarrow \alpha \in]\frac{\pi}{2} + 2k\pi, \frac{3\pi}{2} + 2k\pi[$$

$$if\ \langle w, x\rangle \geq 0\ \Longrightarrow \cos(\alpha) \geq 0 \Longrightarrow \alpha \in [\frac{-\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi]$$

$$(k \in \mathbb{Z})$$

We are going to deal with angles within the range $[0, \pi]$.

$$\begin{cases} if\ \langle w, x\rangle < 0 \Longrightarrow \alpha > \frac{\pi}{2} \\ if\ \langle w, x\rangle \geq 0 \Longrightarrow \alpha \leq \frac{\pi}{2} \end{cases}$$

Perceptron

- If $x \in P \; and \; \langle w, x \rangle < 0 \Longrightarrow$ we should rotate w near to x so that $\alpha \leq 90°$, this is can be done by adding x to w:

$$w_{new} \leftarrow w + x$$

Here:

$$\alpha_{new} < \alpha$$

- If $x \in N \; and \; \langle w, x \rangle \geq 0 \Longrightarrow$ we should rotate w away from x so that $\alpha > 90°$, this is can be done by substructing x from w:

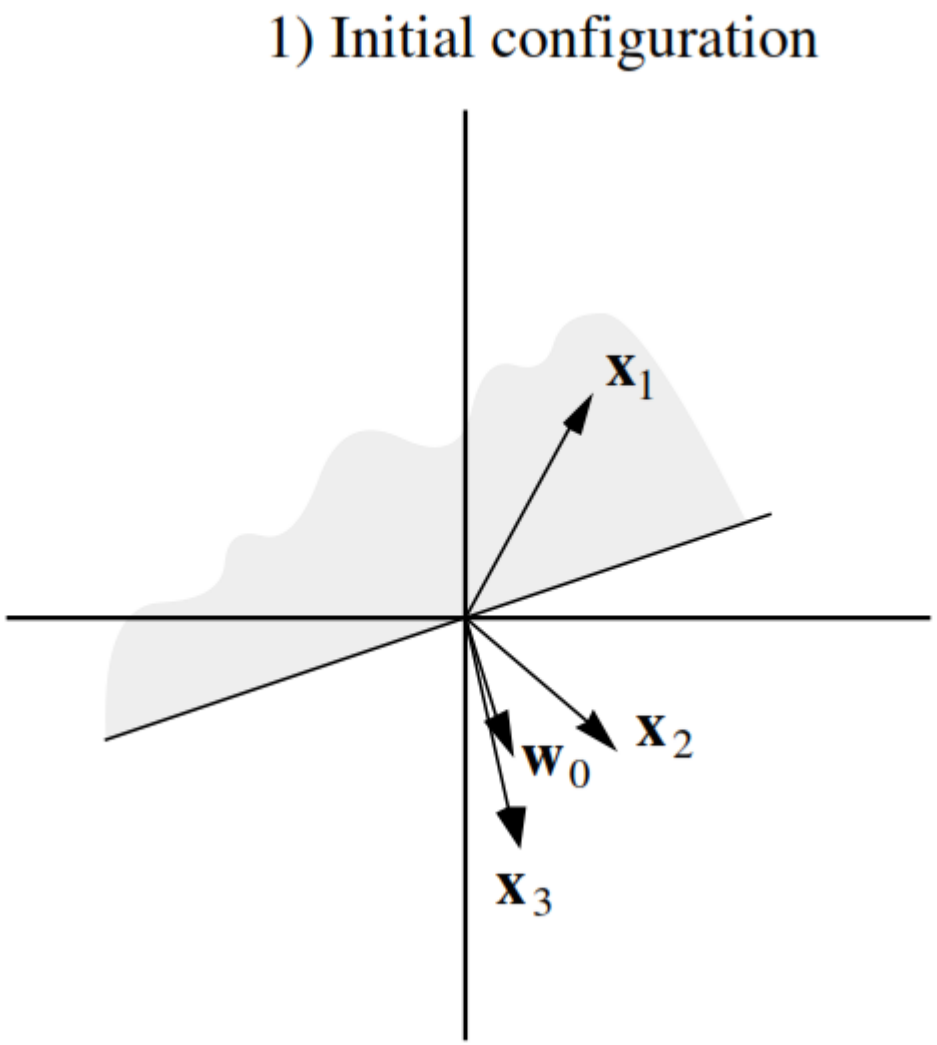
$$w_{new} \leftarrow w - x$$

Here:

$$\alpha_{new} > \alpha$$

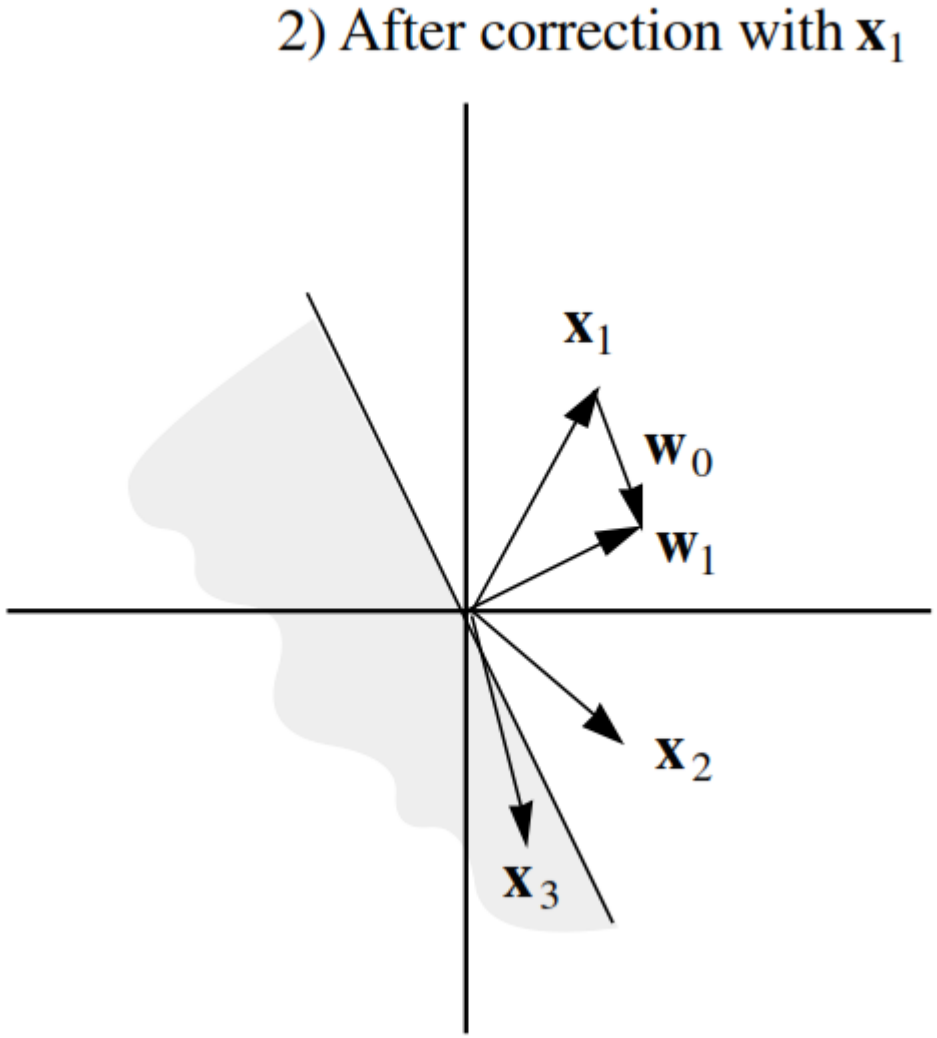
Perceptron

Geometric Visualization:



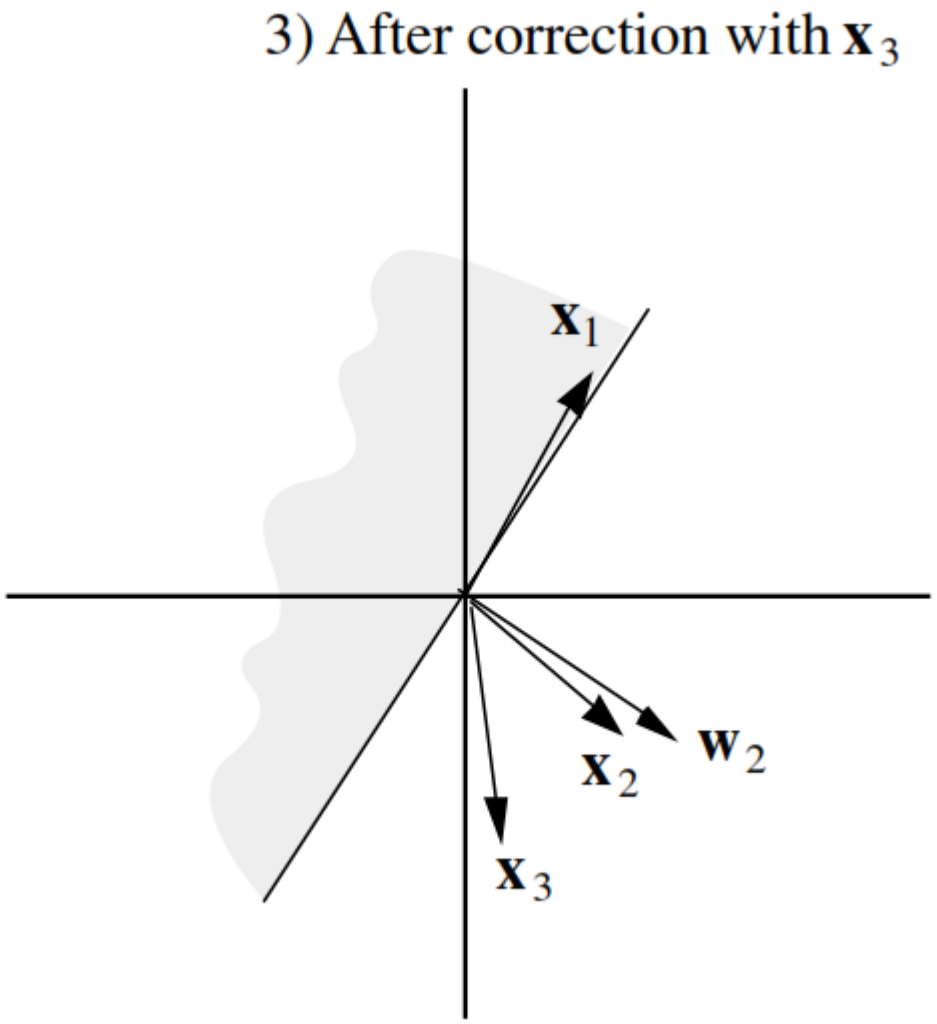
Perceptron

Geometric Visualization:



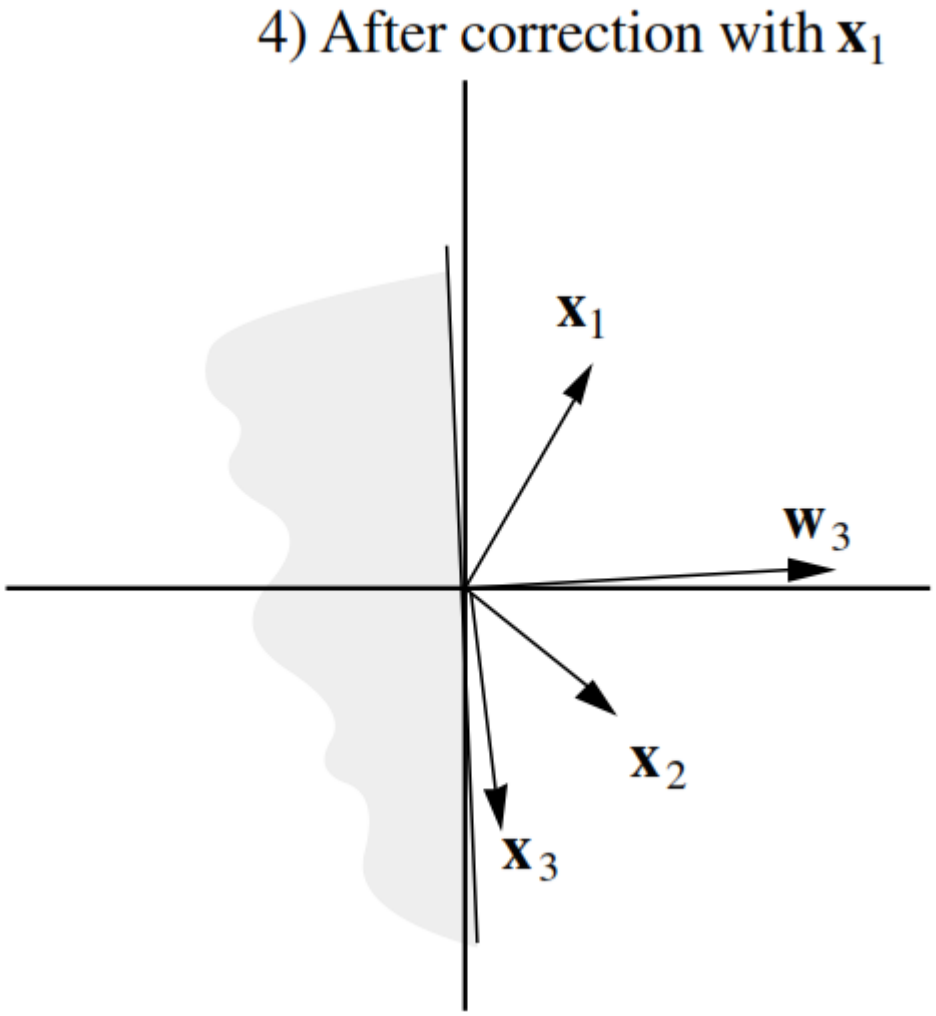
Perceptron

Geometric Visualization:



Perceptron

Geometric Visualization:



Perceptron

Pocket learning algorithm

Input: $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and w_0.
Output: w^*, t and $L_S(w^*)$
Start: $w(0) \leftarrow w_0$
Initialize the weight vector of pocket by the weight vector of PLA.

$$w_s \leftarrow w_0$$

for t = 1 ,…, T_{max} :
  Execute PLA for one weight update to obtain w(t).
  Evaluate $\mathrm{L_S}(w(t))$
  if $\mathrm{L_S}\big(w(t)\big) < \mathrm{L_S}(w_s)$
    $w_s \leftarrow w(t)$
  endif
  Return $\mathrm{w}^* \leftarrow w_s$, t and $L_S(w^*)$
endfor

end

Adaline

Adaline is an improvement of perceptron model developped in 1960 by Widrow and Hoff.
Adaline owns two hypotheses.

- During the training:

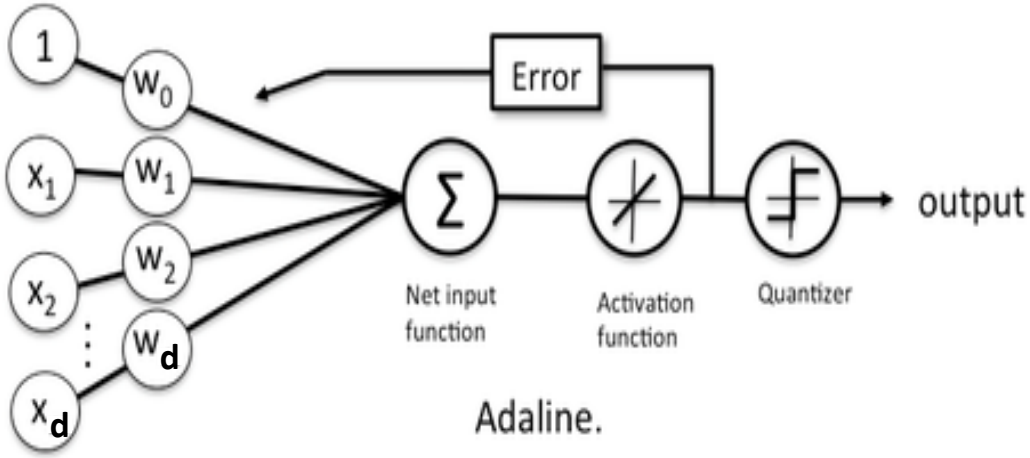
$$\boldsymbol{h_1(x) = \hat{y} = \sum_{i=0}^{d} w_i x_i = w^T x}$$

- After the training:

$$\boldsymbol{h_2(x) = sign(w^T x) = \begin{cases} +1 \text{ si } w^T x \geq 0 \\ -1 \text{ si } w^T x < 0 \end{cases}}$$

Empirical error:

$$L_S(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - w^T x_i)^2$$



Adaline.

Adaline

Delta rule learning algorithm

Input: $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and w_0.
Output: w^*, t and $L_S(w^*)$
Start: $w \leftarrow w_0$

Compute: $L_S(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - w^T x_i)^2$

```
for t = 1 ,…, T_max:
    for i = 1, …, n:
      if (e = y_i − w^T x_i) ! = 0
          w ← w + 2.e.x_i
      endif
    endfor
Endfor
 Return w* ← w , t and L_S(w*)
end
```